



the applications are endless,” he added.

On innovation, Chara emphasised that ferrite magnets are locally feasible: “The challenge is not making ferrite magnets; they are established in India, but designing efficient motors around them.” Pilot programmes are underway with OEMs, validating performance in cost-sensitive segments, with adoption expected over five to ten years.

Meanwhile, in Pune, Conifer Motors is refining iron-based ferrite magnets, which is a more accessible material made from iron oxide that has been manufactured in India for decades. With a proprietary stator design, Conifer has increased torque output to levels comparable to those of rare earth motors. “By rethinking the motor shape and not materials, we can match performance and control output performance”, said the Conifer team.

Conifer, based in Sunnyvale, California, US, with a manufacturing facility in Pune, is developing rare-earth-free motors and powertrains for mobility and industrial applications. “We build motors for two-wheelers, scooters, small EVs, as well as stationary products like fans, pumps, and conveyor belts,” the team said. Their in-wheel powertrain for mobility applications boosts range by up to 30%, reduces weight and size by 15%, and increases continuous torque, all without using rare earth magnets. For stationary motors, efficiency reaches IE5 standards.

Besides startups, several Indian companies are now turning to rare-earth-free technologies. Sterling E-Mobility has partnered with UK-based Advanced Electric Machines (AEM) to locally produce magnet-free traction motors utilising AEM’s high-density reluctance motor (HDRM) platform, which eliminates the use of rare earths while maintaining torque and ensuring recyclability. Pilot production is underway in Faridabad.

Ola Electric, in October 2025, became India’s first two-wheeler OEM to receive AIS-041 certification for a rare-earth-free ferrite motor, available in 7kW and 11kW variants. Tata Elxsi’s ferrite-based motor, first revealed in 2023, could follow once commercial scaling begins.

Final thoughts

The rare earth challenge has become a test of industrial strategy, innovation, and technological foresight. India’s approach so far, focused on exploration, magnet manufacturing incentives, and critical mineral missions, has laid the foundation for resilience. However, true independence seems to depend on how effectively the country channels R&D funding, startup innovation, and circular economy technologies, such as recycling and material substitution. As Applied Materials’ Rengarajan points out, breakthroughs in recovering materials from electronic waste and improving processing efficiency can ease global supply pressures and lower strategic vulnerabilities.

Globally, the path to de-risking remains gradual: immediate efforts will focus on stockpiling and emergency sourcing, followed by mid-term investments in refineries and magnet plants, and finally, long-term diversification of supply chains. However, even by 2040, parity with China’s integrated rare earth ecosystem may remain elusive. For India, thus, the opportunity lies in bypassing China’s model altogether. We can do this through rare-earth-free technologies, sustainable design, and indigenous innovation that make the supply itself optional. **EFY**

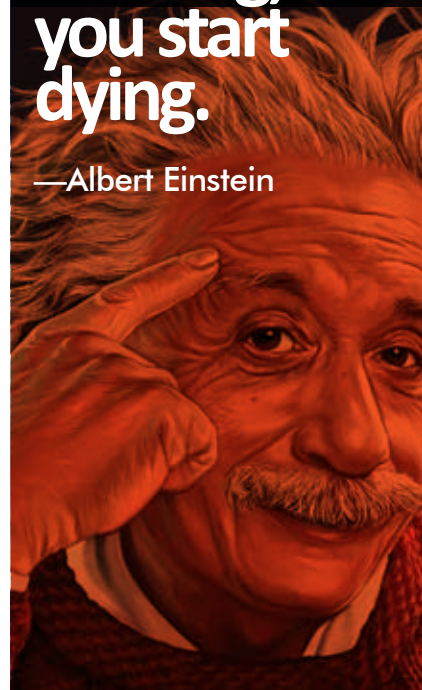
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Once you stop learning, you start dying.

—Albert Einstein



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